

Official Agricultural Extension Advisory: Bacterial Leaf Blight in Rice (Paddy)

Causal Pathogen: Xanthomonas oryzae pv. oryzae

Issuing Source: TNAU Agritech Portal & IRRI Rice Knowledge Bank | Date: 2024-04-10

1. Identification and Morphological Symptoms

Water-soaked lesions begin at leaf margins and tips, rapidly enlarging along the leaf edges with characteristic wavy or undulating borders. Lesions turn straw-yellow to bleached white as the tissue dies. In early morning under moist conditions, milky bacterial ooze droplets form on young lesions, drying into small yellowish beads. In systemic 'kresek' phase (seedling stage), leaves wilt, roll up, and die rapidly.

2. Environmental Epidemiology & Causes

Bacterium enters leaves through natural openings (hydathodes) or mechanical injuries caused by high winds or storms. Promoted by high ambient temperatures (25-34 deg C), high humidity (>85%), wind-driven heavy rains, deep standing water in fields, and heavy nitrogen fertilization.

3. Cultural Management and Field Prevention

Prevention: Use certified disease-free seed and resistant/tolerant cultivars (e.g., IR64, Improved Samba Mahsuri). Avoid clipping seedling leaf tips during transplanting. Avoid deep submergence of rice hills.

Cultural Practices: Adopt intermittent irrigation and field drainage to expose the soil to air. Withhold top dressing of nitrogen fertilizers immediately upon first symptom detection. Increase potassium application (Muriate of Potash) by 25% to enhance host tissue resistance.

4. Organic and Biological Management

Spray 20% fresh cow dung slurry extract (supernatant) twice at 15-day intervals. Seed treatment with *Pseudomonas fluorescens* @ 10 g/kg seed and foliar spray of *P. fluorescens* talc formulation @ 1 kg/ha in 500 liters of water.

5. Verified Chemical Interventions

Seed soaking in Streptocycline (90% Streptomycin sulphate + 10% Tetracycline hydrochloride) @ 0.1 g/L with Copper Oxychloride @ 1.0 g/L for 8-10 hours. Foliar spray of Copper Hydroxide 77% WP @ 2.0 g/L or Streptocycline @ 0.1 g/L + Copper Oxychloride 50% WP @ 2.5 g/L during active tiller stage. Avoid spraying during hot sunny hours.

• **Active Ingredient:** Copper Hydroxide 77% WP | **Reference Formulation:** Kocide 2000 (standard registered formulation) | **Registrant:** Certis / Kocide LLC (approved registrant) | **Source:** DPPQS CIBRC & TNAU Technical Advisory (2024-01-20)

• **Active Ingredient:** Streptomycin sulphate 90% + Tetracycline hydrochloride 10% SP | **Reference Formulation:** Streptocycline | **Registrant:** Hindustan Antibiotics Ltd | **Source:** ICAR-IIRR Technical Manual (2024-03-01)

Notice: Chemical recommendations are derived strictly from central agricultural research advisories. Farmers must consult local district agricultural extension officers for local dosage and label claims.